

Abbreviated CV

Geoffrey K. Vallis
Professor, Ocean Science, UCSC

EDUCATION

1977. B.A., Physics, Oxford University

1981 Ph. D., Physics London University (Imperial College).

PROFESSIONAL HISTORY

1989 – Present. Professor of Marine Science. University of California, Santa Cruz

1983 – 1988. Assistant/ Associate Researcher/ Lecturer. SIO, UCSD.

1981 – 1983. Post-doctoral Researcher, SIO, UCSD.

RESEARCH INTERESTS

Ocean Dynamics and Circulation; Numerical Modelling; Geophysical Fluid Dynamics; Atmospheric Dynamics.

SELECTED RECENT PUBLICATIONS

Vallis, G.K. 1995: Potential vorticity inversion and balanced equations of motion for rapidly rotating and stratified flow. *Quart. J. Roy. Meteor. Soc.* (in press.)

Samelson, R. S. and Vallis, G.K. 1995: Boundary value problems of the planetary geostrophic equations with Rayleigh damping. *J. Phys. Ocean.* (submitted)

Cummins, P. and Vallis, G.K. 1994: Solvers for separable and non-separable elliptic problems in irregular domains. *Assoc. of Comp. Mach., Trans. Mathm. Soft.* 20, No. 3, 247-261.

Warn, T., Bokhove, O., Shepherd, T.G. and Vallis, G.K. 1994: Rossby number expansions, slaving, and balance dynamics. *Quart. J. Roy. Meteor. Soc.* 121, 723-739.

Wang, J. and Vallis, G.K. 1994: Emergence of Fofonoff states in inviscid and viscous ocean circulation models. *J. Mar. Res.* 52, 85-127.

Vallis, G.K. and Maltrud, M.E. 1993: Generation of mean flows and jets on a beta-plane and over topography. *J. Phys. Ocean* 23, 1346-1362.

Sundermyer, M. and Vallis, G.K. 1993: Correlation dimension of primitive equation and balanced models. *J. Atmos. Sci.* 50, 2556-2564.

Maltrud, M. and Vallis, G.K. 1993: Energy and enstrophy transfer in numerical simulations of two-dimensional turbulence. *Physics of Fluids A* 5 1760- 1775.

Vallis, G. K. 1993: Statistical mechanics, turbulence, and ocean circulation. in *Statistical Methods in Physical Oceanography*, Aha Huiakoa proceedings, ed. P. Muller and G. Holloway. 473-491.

Vallis, G.K. 1992: Mechanisms and parameterizations of geostrophic adjustment and a variational approach to balanced flow. *J. Atmos. Sci.* 49, 1144-1160.

Vallis, G.K. 1992: Problems and phenomenology in two-dimensional turbulence. In *Non-linear Phenomena in Oceanic and Atmospheric Dynamics*. eds. G. Carnevale and R. Pierrehumbert. 1-25. Springer-Verlag. (refereed chapter in book)

Maltrud, M. and Vallis, G.K. 1991: Energy spectra and coherent structures in forced two-dimensional and geostrophic turbulence. *J. Fluid Mech.* 228, 321-342.

- Auad, G., Pares-Sierra, A., and Vallis, G.K. 1991: Energetics and diagnostics of a model of the circulation in the California Current System. *J. Phys. Ocean*, 21, 1534-1552.
- Carnevale, G.F. and Vallis, G.K. 1990: Pseudo-advective relaxation to stable two-dimensional states. *J. Fluid. Mech.*, 213, 549-571.
- Vallis, G.K., Carnevale, G.F. and Shepherd, T.G. 1990: A natural method for the stable states of Hamiltonian systems. In *Topological Fluid Mechanics, Proceedings of the IUTAM Symposium*. ed. H.K. Moffatt and A. Tsinober, 429-439. (refereed conference proceedings)
- Carnevale, G.F. and Vallis, G.K. 1990: Iso-vortical energy variation in two-dimensional flows. In *Topological Fluid Mechanics, Proceedings of the IUTAM Symposium*. ed. H.K. Moffatt and A. Tsinober, 294-303. (refereed conference proceedings)
- Carnevale, G.F., Purini, R., Briscolini, M., and Vallis, G.K. 1989: Influence of topography on modon propagation and survival. In *Mesoscale/Synoptic Coherent Structures in Geophysical Turbulence*, ed. J.C.J. Nihoul and B.M. Jamart. Elsevier Science Publishers. (review article)
- Pares-Sierra, A., and Vallis, G.K. 1989: A fast semi-direct method for the numerical solution of non-separable elliptic equations in irregular domains. *J. Comp. Physics*. 82, 398-412.
- Vallis, G.K., Carnevale, G.F. and Young, W.R. 1989: Extremal energy properties and construction of stable solutions of the Euler equations. *J. Fluid. Mech.* 203, 133-152.
- Vallis, G.K. and Hua, B.-L. 1988: Eddy viscosity of the anticipated potential vorticity method. *J. Atmos. Sci.* 45, 617-627.
- Carnevale, G. F., Vallis, G.K., Purini, R., and Briscolini, M. 1988: Propagation of barotropic modons over topography. *Geophys. Astrophys. Fluid Dynam.* 41, 45-101.
- Vallis, G.K. 1988: A numerical study of transport properties in eddy resolving and parameterized models. *Quart. J. Roy. Meteor. Soc.*, 114, 183-204.
- Carnevale, G.F., Vallis, G.K., Purini, R., and Briscolini, M. 1988: The role of initial conditions in flow stability, with applications to modons. *Phys. Fluids*, 31(9), 2567-2572.
- Carnevale, G.F., Purini, R., Briscolini, M., and Vallis, G.K. 1988: Numerical experiments on modon stability to topographic perturbations. *Phys. Fluids*. 31(9), 2562-2567.
- Vallis, G.K. 1988: Conceptual models of El Niño and the Southern Oscillation *J. Geophys. Res. (Oceans)* 93, 13979-13991.
- Vallis, G.K. and Roads, J.O. 1986: Turbulent effects in large scale flow over topography. In *Proceedings of Second International Symposium on Tibet Plateau and Mountain Meteorology*, Beijing. (eds. Z. Baozhen and E. Reiter) pp 390-407, Academia Sinica, China. (conference proceedings)
- Vallis, G.K., 1986: El Niño: a chaotic dynamical system? *Science* 232, 243-245.
- Vallis, G.K. 1985: On the spectral integration of the quasi-geostrophic equations for doubly-periodic and channel flow. *J. Atmos. Sci.* 42, 95-99.
- Vallis, G.K. 1985: Remarks on the predictability properties of two and three dimensional flow. *Quart. J. Roy Meteor. Soc.*, 111, 1039-1047.
- Vallis, G.K. 1985: Instability and flow over topography. *Geophys. Astrophys. Fluid Dyn.*, 34, 1-38.
- Roads, J.O. and Vallis, G.K. 1984: An energy balance model with cloud feedbacks. *Tellus*, 36A, 236-250.
- Roads, J.O., Vallis, G.K. and Remer, L. 1984: Cloud/climate sensitivity experiments. In "Climate Processes and Climate Sensitivity," 92-107. *Geophysical Monograph* 29, ed J. Hansen and

- T. Takahashi, American Geophysical Union. (refereed article in book).
- Vallis, G.K. and Roads, J.O. 1984: Large-scale stationary and turbulent flow over topography. *J. Atmos. Sci.* 41, 3255-3271.
- Vallis, G.K. 1983: On the predictability of quasigeostrophic flow: the effects of beta and baroclinicity. *J. Atmos. Sci.*, 40, 10-27.
- Vallis, G.K. 1983: Barotropic and baroclinic predictability in geostrophic turbulence. *Am. Inst. Physics Proceedings*, "The Predictability of Fluid Motions." 117-133, ed. G. Holloway and B. West. (conference proceedings)
- Carnevale, G.F. and Vallis, G.K. 1983: Applications of entropy to predictability theory. *Am. Inst. Physics Proceedings*, "The Predictability of Fluid Motions." 577-593, ed. G. Holloway and B. West. (conference proceedings)
- Vallis, G.K. 1982: A statistical dynamical climate model with a simple hydrology cycle. *Tellus*, 34, 211-227.
- Cayan, D., Harnack, R. and Vallis, G.K. 1982: Report on workshop on weather and climate variability during winter 1981-1982. *Bull. Am. Met. Soc.*, 63, 1079-1183.